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Case-by-case: Emotional stimulus significance and the modulation of the EPN and LPP

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Abstract

Stimuli encountered in the environment are continuously evaluated according to their affective stimulus significance. Numerous event-related potential studies have shown that the early posterior negativity (EPN) and the late positive potential (LPP) are larger for pictures high in emotional arousal compared to low arousing images. Here, we used a case-by-case approach to determine the emotional modulation of the EPN and LPP at the level of the individual case. We probed the effect in three studies regarding behavior systems of predator fear, disease avoidance, and sexual reproduction. In each study, high and low arousing pictures from the respective behavior system were presented and event-related brain potentials measured. At the group level, previous findings regarding an enhanced EPN and LPP for high compared to low arousing stimuli were replicated. Novel findings were observed when analyzing each case individually. The majority of participants showed significantly larger EPN and LPP amplitudes to pictures showing threatening compared to harmless animals (15 and 16 out of 16 tests, respectively), images of disease and mutilation compared to neutral people (15 and 17 out of 18 tests, respectively), and high-arousing erotic compared to romantic pictures (17 and 17 out of 17 tests, respectively). Further analyses exploring the tradeoff between specificity and sensitivity suggest reliance on picture categories of sexual reproduction and disease avoidance. Overall, the present findings demonstrate that the case-by-case approach can be useful for revealing a general principle of emotional stimulus processing common-to-all. Implications for the reproducibility of the findings and biomarker development are discussed.

1 | INTRODUCTION

The fast and reliable detection of positive and negative stimuli facilitates adaptive behavior (Lang et al., 1997; Öhman et al., 2000). Functional and evolutionary considerations suggest that stimuli encountered in the environment are evaluated according to their affective stimulus significance. Numerous event-related potential studies on emotional picture processing revealed that two ERP components are modulated by the intensity of picture arousal (Hajcak et al., 2010; Lang & Bradley, 2010; Schupp et al., 2006). Specifically, an early posterior negativity (EPN), between 150 and 350 ms, and a late positive potential (LPP), between 350 and 700 ms post-stimulus onset, are larger for emotionally arousing as compared to low arousing, control stimuli (Cuthbert et al., 2000; Junghöfer et al., 2001; Schupp, Flaisch, et al., 2006; Schupp et al., 2003). Accordingly, it has been suggested that the EPN and LPP components represent an indicator of the affective significance of the stimuli (Schupp, Flaisch, et al., 2006). The present research series expands on these findings by using a

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case-by-case approach to assess the process of affective stimulus evaluation at the level of the individual case.

Previous research demonstrated the emotional modulation of EPN and LPP components across a range of emotional stimuli and experimental conditions, thus, demonstrating the generality of this phenomenon. Stimulus categories included stimuli from the International Affective Picture System (e.g., Schupp et al., 2004), emotional facial expressions (Schindler & Bublatzky, 2020), emotional words (Schindler & Kissler, 2016), symbolic gestures (Flaisch et al., 2011), clashing moral statements (Van Berkum et al., 2009), cues predicting painful stimulation to the partner (Wu et al., 2017), and personal trait feedback provided either by a human sender or computer (Schindler et al., 2015). Furthermore, the phenomenon has been observed for varying exposure times (Cuthbert et al., 2000; Junghöfer et al., 2001; Peyk et al., 2009; Schupp, Flaisch, et al., 2006) and shown to be stable across stimulus repetition (Codispoti et al., 2007; Ferrari et al., 2020; Schupp, Stockburger, et al., 2006). These findings support the notion that the EPN and LPP may serve as a neural indicator of the processing of emotionally arousing stimuli.

However, different experimental approaches can be used to determine general principles of emotion processing. One approach draws inferences by studying the effect at the group level (Danziger, 1990; Lamiell, 1998; Robinson, 2011). Calculating group statistics, such as dependent *t* tests and repeated-measure analysis of variance, the numerous ERP studies on emotional stimulus processing relied on the group approach to reveal significant effects (Hajcak et al., 2010; Schindler & Bublatzky, 2020; Schupp, Flaisch, et al., 2006). Thus, the empirical regularity of increased EPN and LPP components to high arousing emotional stimuli is referring to the group as an aggregate whole, that is, a "hypothetical person." A complementary approach to reveal general principles of emotion processing is to study the effect at the level of the individual person. Thus, the case-by-case approach can reveal principles of emotion processing that are common to all individuals in the research sample (Danziger, 1990; Lamiell, 1998; Robinson, 2011). Determining enhanced EPN and LPP components to emotionally significant stimuli for the individual case can reveal how many individuals show the effect. This information is needed to provide strong evidence for a general principle of emotion functioning common-to-all, potentially enabling the interpretation of individual cases and translation to the clinical domain.

A functional-evolutionary perspective on emotions was adopted in order to select stimulus materials. Distinct emotional behavior systems have evolved to accomplish critical tasks for survival and recreation which are tuned to selectively respond to certain prototypical stimuli (Domjan, 1994; Fanselow, 1994; Öhman et al., 2000; Timberlake, 2001). Selecting stimulus materials related to behavior systems allows to select prototypical high arousing stimuli with shared emotional significance across individuals. Furthermore, low arousing control stimuli can be chosen which are similar in terms of content category, figure-ground composition, and low-level physical stimulus characteristics (cf. Bradley et al., 2007; Junghöfer et al., 2001). Finally, it can be informative to compare EPN and LPP effects across different behavior systems, allowing to determine the generality of the observed effects and developing robust experimental paradigms.

In the present research series, we used the case-by-case approach to determine whether the process of affective stimulus evaluation assessed by the EPN and LPP component can be reliably revealed at the level of the individual case. Toward this end, Studies 1–3 separately investigated stimuli from the behavior systems of predator fear, disease avoidance, and sexual reproduction in a free viewing task. Specifically, bootstrap analyses were used to determine statistically significant effects within each individual case (Efron & Tibshirani, 1993; Wasserman & Bockenholt, 1989). Group analyses served to replicate previous findings and to determine the topography of the EPN and LPP sensor clusters used for the case-by-case analysis. We hypothesized that the effect can be assessed at the level of the individual case and determined the proportion of cases showing increased EPN and LPP components to high as compared to low emotionally arousing pictures. The observed case-by-case replication within studies indicates whether the examined effect is common-to-all and the systematic replication across studies reveals the robustness and generality of the findings. To complement the focus on sensitivity, that is, how many individuals show the supposed EPN and LPP effect, we conducted further analyses to assess the specificity of the effect, that is, how many individuals show no effect when no significant effect is expected.

2 | METHOD

2.1 | Participants

All participants were students recruited from the University of Konstanz. The final sample of Study 1 included 16 participants (8 female) between 18 and 32 years of age ($M = 22.8$, $SD = 4.1$). Data from one additional participant were discarded due to an overtired and exhausted behavioral state during the experiment with visible signs of sleepiness. Study 2 included 18 participants (9 female) between 19 and 33 years of age ($M = 23.4$, $SD = 3.3$). Study 3 included 17 females between 19 and 25 years of age ($M = 21.1$, $SD = 1.8$).

All participants were healthy at the time of testing and reported no history of neurological or psychiatric disorders. Vision was normal or corrected-to-normal. Participants received monetary compensation or course credit for participation. The ethical committee of the University of Konstanz approved the experimental procedure in accordance with the regulations of the Declaration of Helsinki, and all methods were carried out in full compliance with the approved guidelines. All participants provided informed consent and were debriefed after the experiment.

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2.2 | Stimuli

In Study 1, we examined the predator fear system and the stimulus materials comprised two categories of animal pictures (Öhman et al., 2000). The high arousing category comprised 10 pictures of wild animals in dangerous, threatening poses (e.g., tiger, shark, and alligator). The low arousing category included 10 pictures of harmless animals in non-threatening poses (e.g., cat, sheep, and lizard).

Study 2 examined disease avoidance as a different kind of defense against potentially harmful stimuli (Neuberg et al., 2011). The sight of bleeding or injured and/or deformed human bodies represents a prototypical stimulus class for engaging the disease avoidance system (Marks, 1988). Accordingly, the high arousing emotion category comprised 10 pictures of mutilated and injured bodies, which was compared to 10 pictures showing uninjured humans in neutral poses.

Focusing on a positive-valence behavior system, Study 3 selected stimuli related to sexual reproduction (Domjan, 1994). The high arousing emotion category comprised 10 pictures showing couples in an explicit erotic posture compared to 10 pictures showing couples in a romantic pose, that is, hugging or kissing (low arousing category).

Twenty pictures were selected for each of the three behavior systems. The pictures for the strong ($N = 10$) and weak ($N = 10$) emotional categories were selected to be similar in overall composition and were standardized with respect to brightness in the red, green, and blue channel in order to minimize physical stimulus differences between categories. Furthermore, pictures were presented in the original as well as horizontally flipped direction to control for horizontal eye movements, resulting in 20 pictures per category.

Arousal and valence ratings of the pictures, obtained after EEG measurements using the Self-Assessment Manikin (Bradley & Lang, 1994), were in line with normative ratings. Specifically, regarding arousal, threatening ($M = 5.82$, $SD = 2.05$) compared to safe animals ($M = 2.26$, $SD = 0.94$), mutilations ($M = 5.92$, $SD = 1.71$) compared to neutral people ($M = 2.91$, $SD = 1.06$), and erotica ($M = 5.71$, $SD = 1.05$) compared to romantic pictures ($M = 2.58$, $SD = 1.04$) were evaluated as more arousing, $t_s \geq 8.0$, $p < .001$. Regarding valence, threatening ($M = 3.46$, $SD = 1.14$) compared to safe animals ($M = 6.86$, $SD = 1.05$), and mutilations ($M = 2.52$, $SD = 0.85$) compared to neutral people ($M = 6.24$, $SD = 0.70$) were rated as more unpleasant, $t_s \geq 8.4$, $p < .001$. In addition, romantic images ($M = 6.58$, $SD = 0.91$) were evaluated as more pleasant compared to erotic pictures ($M = 5.39$, $SD = 1.32$), $t(15) = 2.4$, $p < .05$).

2.3 | Experimental design

Varying picture contents across studies, other parameters of the experimental design were identical across studies. Pictures were displayed for 150 ms, preceded by a fixation cross shown for 100 ms. The intertrial interval varied from 617 to 967 ms ($M = 790$ ms). Previous studies have revealed little evidence for the habituation of emotion processing in terms of EPN and LPP modulation (Ferrari et al., 2020; Schupp, Stockburger, et al., 2006). Accordingly, the picture set was repeated 45 times, resulting in a total of 1,800 presentations per experiment. The participants viewed the pictures from the two categories in a pseudo-randomized order for each participant. No more than three consecutive presentations of the same picture category were allowed, and transition frequencies between picture categories were controlled. Participants were instructed to simply view the images.

EEG data recording lasted approximately 40 min and included two short breaks (~5 min) for posture adjustment and impedance checks.

2.4 | EEG data acquisition and analysis

EEG data acquisition was identical for Studies 1–3. Brain and ocular scalp potentials were measured with a 256-lead geodesic sensor net (HCGSN), filtered on-line below 100 Hz (DC), and sampled at 250 Hz using NetStation acquisition software and EGI amplifiers (*Electrical Geodesics Inc., Eugene, OR*). Electrode impedance was kept below 40 kΩ, as recommended for this type of electroencephalogram (EEG) amplifier by EGI guidelines. Data were recorded continuously using the vertex sensor as a reference electrode. A 40-Hz digital low pass filter was applied off-line to the continuous EEG data. The data were then corrected for ocular artifacts based on a multiple regression method, converted to an average reference, and baseline-adjusted (100 ms pre-stimulus) using EMEGS software (Peyk et al., 2011). Artifact rejection was performed based on an elaborate method for statistical control of artifacts, specifically tailored for the analyses of dense sensor EEG recordings (Junghöfer et al., 2000). To ensure equal representation of high and low arousing trials in the bootstrap analyses, trial numbers after artifact rejection were equated between condition by keeping the desired number of trials from the condition with more trials based on a vector of randomly permuted integers. Finally, separate average waveforms were calculated for each individual and the two picture categories (low and high arousal).

2.5 | Statistical analysis

2.5.1 | Group analysis

Findings from single sensor waveform analysis and visual inspection were used in concert to select bilaterally symmetrical sensor clusters and time windows to score the

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EPN and LPP amplitudes used in the case-by-case analysis and the reporting of the effect in group analysis. Single sensor waveform analysis with factors of emotional arousal (low and high) and behavior system (predator fear, disease avoidance, and sexual reproduction) revealed significant EPN and LPP effects with expected topographies and latencies, reaching significance when adjusting *p*-level for multiple testing. Accordingly, the EPN was scored in an occipito-parietal sensor cluster comprising the following sensors: 106, 107, 108, 113, 114, 115, 116, 117, 121, 122, 123, 124, 125, 126, 133, 134, 135, 136, 137, 138, 139, 145, 146, 147, 148, 149, 150, 151, 156, 157, 158, 159, 160, 165, 166, 167, 168, 169, 174, 175, and 176. The LPP was scored in a centro-parietal cluster including the following sensors: 6, 7, 8, 9, 15, 16, 17, 23, 24, 30, 42, 43, 44, 45, 51, 52, 53, 59, 60, 79, 80, 81, 89, 130, 131, 132, 143, 144, 155, 183, 184, 185, 186, 196, 197, 198, 206, 207, 215, and 257. Mean activity of the EPN and LPP sensor clusters were derived from time windows of 240–300 ms and 380–480 ms, respectively.

To replicate larger EPN and LPP amplitudes for each category of emotionally arousing pictures at the group level, *t* tests were conducted separately for each study.

2.5.2 | Case-by-case analysis

To determine the significance of the measured ERP difference at the individual level, the EPN and LPP data from each participant were submitted to a bootstrap analysis (Efron & Tibshirani, 1993; Wasserman & Bockenholt, 1989). This approach is well-established in research on lie detection, in which individual cases are classified in terms of guilt and innocence (Farwell & Donchin, 1991; Rosenfeld, 2020). Specifically, with 50.000 bootstrap repetitions, each case's EPN/LPP mean data were resampled by randomly (re-)assigning a case's trials to the high and low arousing category (drawn with replacement), and then, calculating the mean difference. Significance ($p < .05$, one-sided) on the individual case level was determined as the proportion of results in the resulting bootstrap distribution that were equal or more extreme than the de facto measured EPN/LPP difference. *p*-values of $p < .05$, $p < .01$, $p < .001$, and $p < .00002$, respectively, indicate that less than 2.500, 500, 50, and 0 out of 50.000 randomized calculations resulted in a difference that was equal or more extreme than the actual observed result (cf. Figure 2). These analyses were conducted using the sensor clusters derived from group analyses to score the EPN and LPP. However, the latency of the effect was allowed to vary between participants, emotion categories, and studies to acknowledge inter-individual variability in functional brain organization. Toward this end, a custom software determined the time window showing the maximum EPN and LPP difference effect (high – low arousal) for each case. The effects were defined to appear within a time window of 150–350 ms for the EPN and 350–750 ms for the LPP component. Using a sliding window algorithm, the mean activity in the EPN cluster across a 60 ms time window and the LPP cluster across a 100 ms time windows were calculated within the respective time boundaries of the EPN and LPP. The time window showing the maximum negative (EPN) or positive (LPP) difference was afterwards selected for each individual case and used for the calculation of the single subject bootstrap distribution.

2.5.3 | Specificity analysis

Information on sensitivity, that is, number of cases showing the predicted effect, is complemented by analysis of specificity revealing the number of nonsignificant cases when no effect is expected. Toward this end, we capitalized on left- and right mirrored presentation of the stimulus pictures. No significant EPN and LPP difference is predicted for picture categories differing only in orientation of the picture contents. Specifically, for each study, the high and low arousing picture categories were analyzed separately splitting the picture contents into left and right picture conditions. The bootstrap analyses were based on scoring the two ERP components with the same sensor clusters and latency parameters as used in the main analysis. As bootstrap analyses were conducted for the two possible directions, that is, left > right; right > left, the number of tests in the specificity analysis is four times the number of the sensitivity analysis. The outcome of the specificity analysis reveals the percentage of tests, when no difference is expected, and the percentage of cases showing false positive effects (false positive rate: 100-specificity).

2.5.4 | Control analyses

Several control analyses were undertaken to determine how the effects varied as a function of different approaches to score the EPN and LPP component.

A first control analyses repeated the main analysis using the same sensor clusters and time windows as used for the group analyses.

In further control analyses, an extended time window was used to score the components. Specifically, the maximum EPN and LPP difference (high – low arousal) was scored in time windows of 120 and 200 ms, respectively, doubling the duration of the time windows from the main analysis.

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3 | RESULTS

3.1 | Group analyses

As shown in Figures 1 and 2, the group analysis replicated previous findings regarding the emotional modulation of the EPN and LPP components for the three behavior systems examined in Studies 1–3. Specifically, a negative difference emerged over posterior sensor sites between 150 and 350 ms for high arousing emotional stimuli. Furthermore, subsequent to the EPN, a relative positivity was observed over centro-parietal sensor sites for high arousing emotional stimuli. Statistical analysis corroborated these impressions for all three studies.

For the EPN, a significant difference was observed for predator fear (Study 1), $t(15) = -6.09$, $p < .001$, Cohen's $d = -1.52$, disease avoidance (Study 2), $t(17) = -6.43$, $p < .001$, Cohen's $d = -1.52$, and sexual reproduction (Study 3), $t(16)$

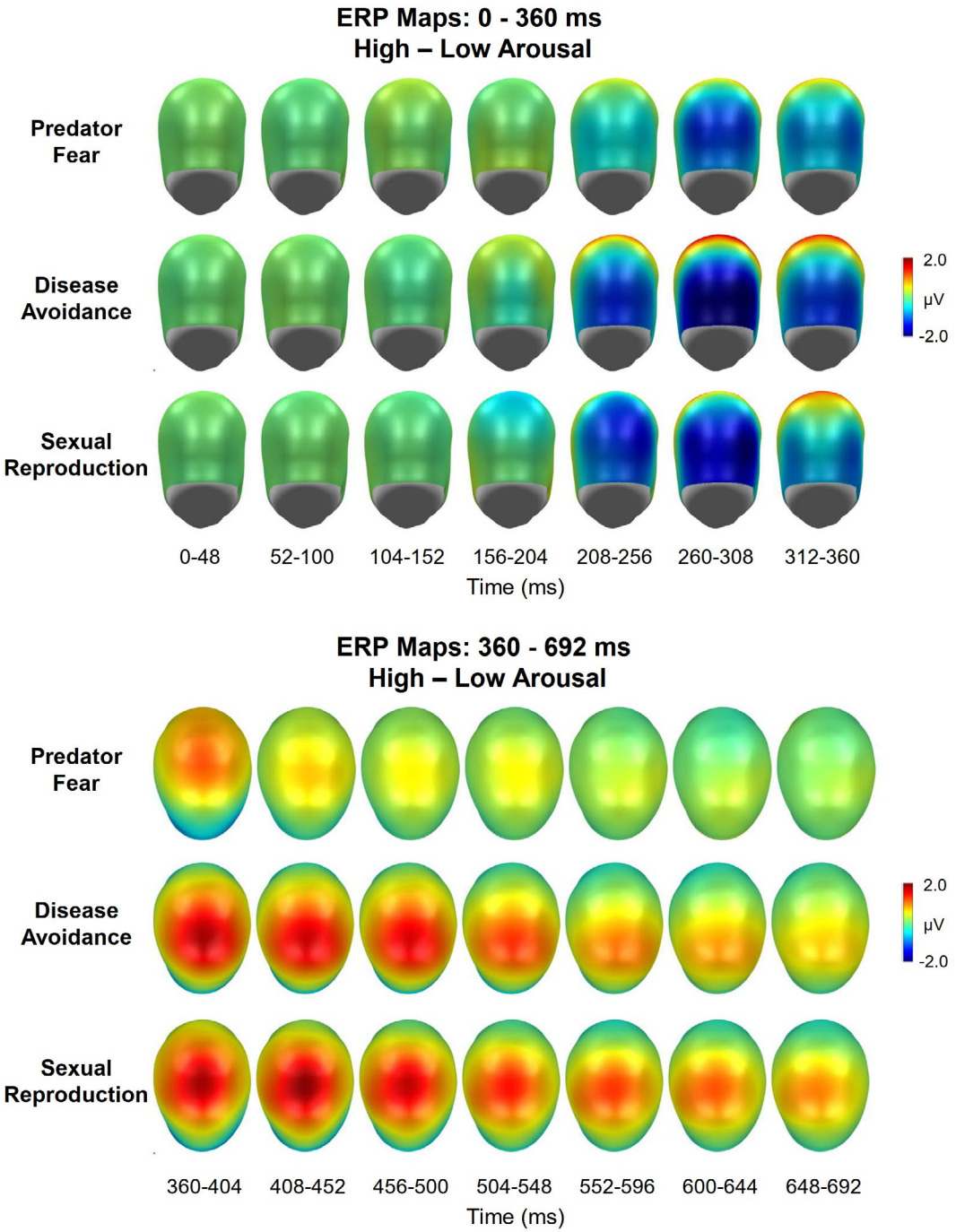

FIGURE 1 Upper panel: Illustration of the EPN effect emerging around 150 ms after stimulus onset and lasting until around 350 ms. A back view of the model head is displayed. Lower panel: Illustration of the centro-parietal LPP effect observed between 350 and 700 ms. A top view is displayed

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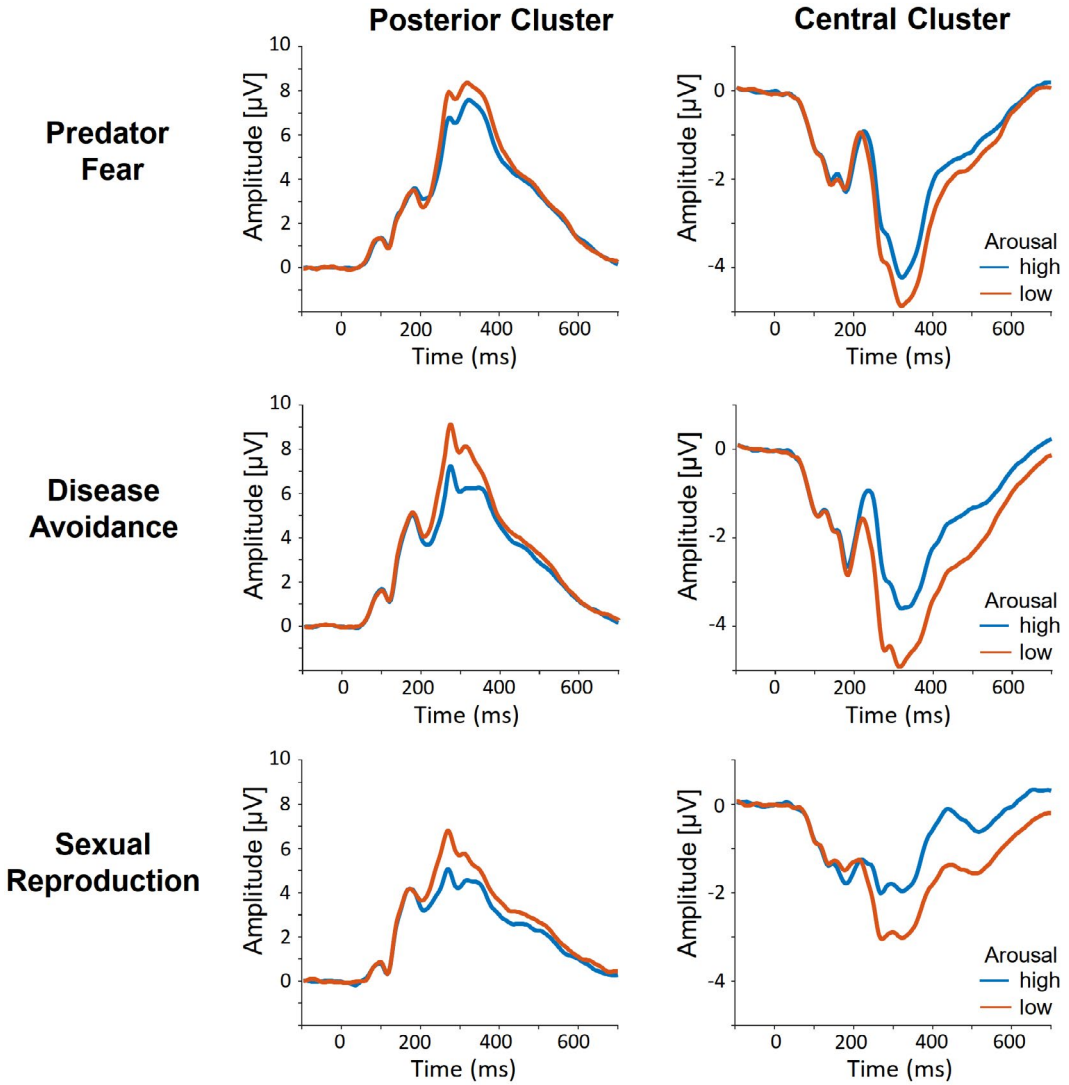

FIGURE 2 Illustration of the ERP waveforms separately for predator fear, disease avoidance, and sexual reproduction and low and high arousal picture categories. Waveforms represent the average across all channels in the sensor cluster used for ERP analysis. The left panels illustrate the EPN effect, appearing as a relative negative shift for high arousing images. The right panels illustrate the LPP effect, appearing as a relative positive shift for high arousing images

= −7.82, $p < .001$, Cohen's $d = -1.90$. Similarly, significant effects were seen in the group analysis of the LPP for predator fear, $t(15) = 8.16$, $p < .001$, Cohen's $d = 2.04$, disease avoidance, $t(17) = 7.94$, $p < .001$, Cohen's $d = 1.87$, and sexual reproduction, $t(16) = 8.96$, $p < .001$, Cohen's $d = 2.17$.

3.2 | Case-by-case analyses

Figure 3 displays the single case bootstrap findings of Studies 1–3 for the EPN and LPP components. The figure shows how many individual cases show a significant effect in accordance with the group effect. Regarding predator fear (Study 1), 15 and 16 out of 16 participants showed significantly larger EPN and LPP amplitudes to pictures showing threatening rather than harmless animals, respectively. For disease avoidance (Study 2), images of disease and mutilation yielded significantly larger EPN and LPP amplitudes relative to images of uninjured humans in 15 and 17 out of 18 participants, respectively. Regarding sexual reproduction (Study 3), all 17 participants showed significantly larger EPN as well as LPP amplitudes to erotica than to romantic pictures. Furthermore, the maximum emotion difference (high – low arousal) detected by the automatic scoring algorithm indicated similar onset latencies of the selected time windows for predator fear (EPN: $M = 257.0$, $SD = 20.42$; LPP: $M = 373.25$, $SD = 44.20$), disease avoidance (EPN: $M = 259.3$, $SD = 17.26$; LPP: $M = 394.00$, $SD = 56.97$), and sexual reproduction (EPN: 241.2, $SD = 17.48$; LPP: $M = 385.4$, $SD = 29.96$). Overall, the findings of Studies 1–3 suggest a high reproducibility of the EPN and LPP effect at the level of the individual case.

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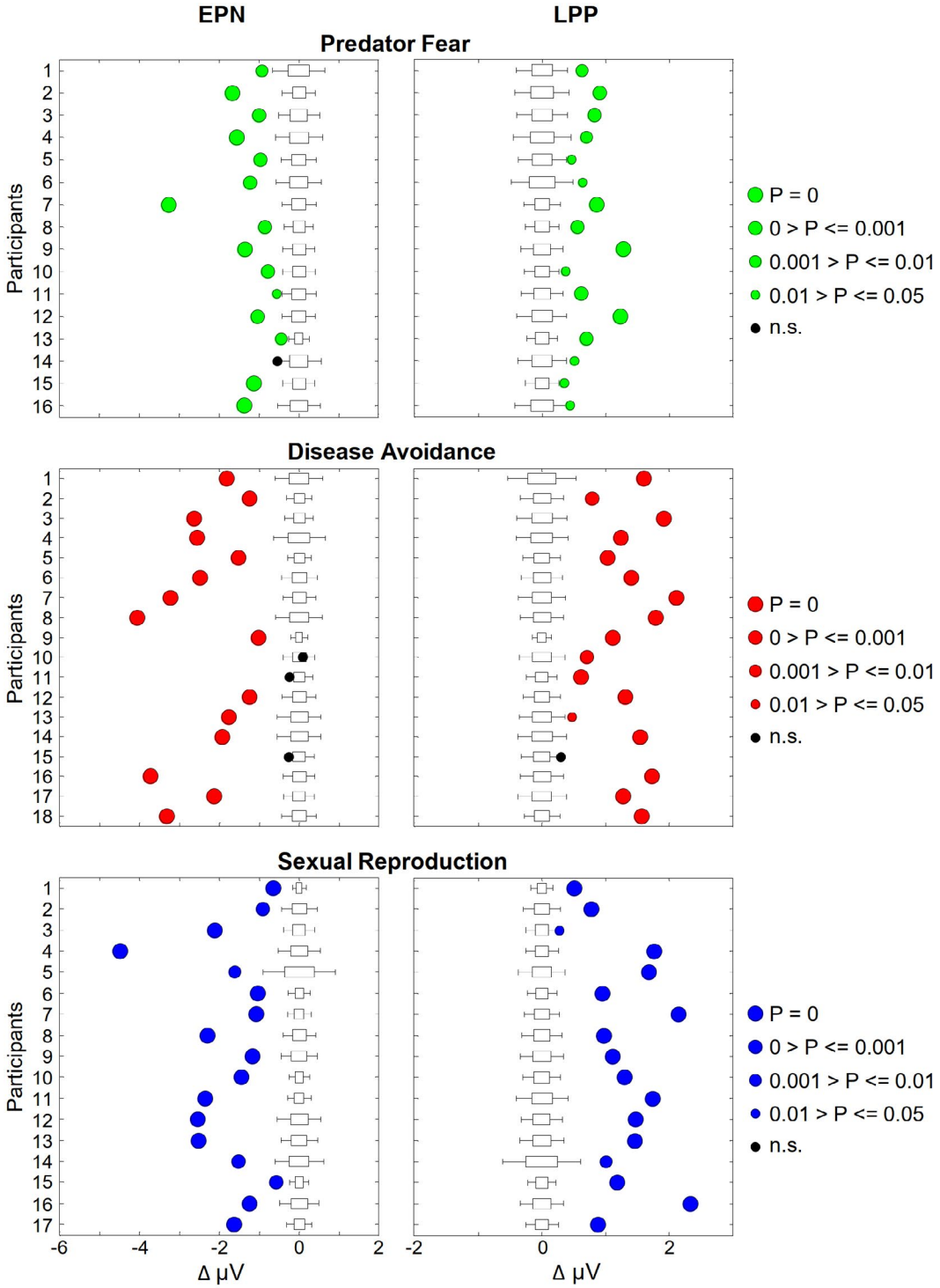

FIGURE 3 Outcome of the case-by-case statistics for the EPN (left) and LPP (right) components in Study 1–3. For each case, a boxplot displays the bootstrap distribution with the bottom and top edges of the box indicating the 25th and 75th percentiles and the whiskers indicating the 5th and 95th percentiles. The dots indicate the de facto measured amplitude difference (high – low arousal). Dots outside the range indicated by the whiskers represent significant effects ($p < .05$) with size and color indicating different p-levels

3.3 | Sensitivity and specificity

Borrowing concepts from diagnostic testing (Button et al., 2013; Kapur et al., 2012; Woo et al., 2017), the performance of a case-by-case analysis should not only be scrutinized in terms how many individuals show the supposed EPN and LPP effect, that is, sensitivity, but also with regard to how many individuals show no effect when no significant effect is expected, that is, specificity. Accordingly, the high and low arousing picture sets for each of the three studies

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were sorted separately in two picture categories showing left-right mirrored pictures equated for picture content for which no significant EPN and LPP effects were predicted.

Relying on $p < .05$ criteria, specificity of the tests ranged between 81% and 92% of the tests for the EPN and LPP components in the three emotion categories. Thus, between 8% and 19% of the tests were significant when nonsignificant effects were predicted, that is, false positive rate (see Figure 4). Further analyses revealed that the magnitude of the effect differed markedly between false positives in the specificity analyses and the emotional modulation of the EPN and LPP in the sensitivity analyses. Specifically, the amplitude difference between conditions in the tests reaching significance was on average about two times larger for the emotional EPN difference compared to the orientation effect in the EPN time window (predator fear: $M = -1.56$ vs. $M = 0.77$; mutilation disease: $M = -2.73$ vs. $M = -0.85$; sexual reproduction: $M = -1.68$ vs. 0.82). A similar amplitude difference was observed also for the LPP (predator fear: $M = 1.07$ vs. $M = 0.56$; disease avoidance: $M = 1.44$ vs. $M = 0.78$; sexual reproduction: $M = 1.27$ vs. $M = 0.65$). Furthermore, the tradeoff between sensitivity and sensitivity is shown in Figure 4 as a function of different p-levels to determine statistical significance. Noteworthy, the rate of "false positives" drops below 10% in all three emotion categories using the $p < .01$ criteria and is negligible at more stringent p-levels. While the EPN and LPP effects remain at more stringent p-level criteria relatively high for categories of disease avoidance and sexual reproduction, the number of significant cases is reduced for predator fear at stronger p-criteria's, in particular for the LPP. Accordingly, while choosing the p-level criteria depends on the specific aim, the tradeoff between specificity and sensitivity for categories of sexual reproduction and disease avoidance suggests the reliance on stronger p-level criteria, that is, $p < .01$.

3.4 | Control analyses

In the main analysis, EPN and LPP were scored in single subject analysis using the same sensor clusters for each study derived from group analysis. While this scoring approach assures a common topography of the ERP components, the latency was allowed to vary in time between individuals within the usual time window of both components. To provide a more conservative approach, a control analyses was undertaken based on the same time windows

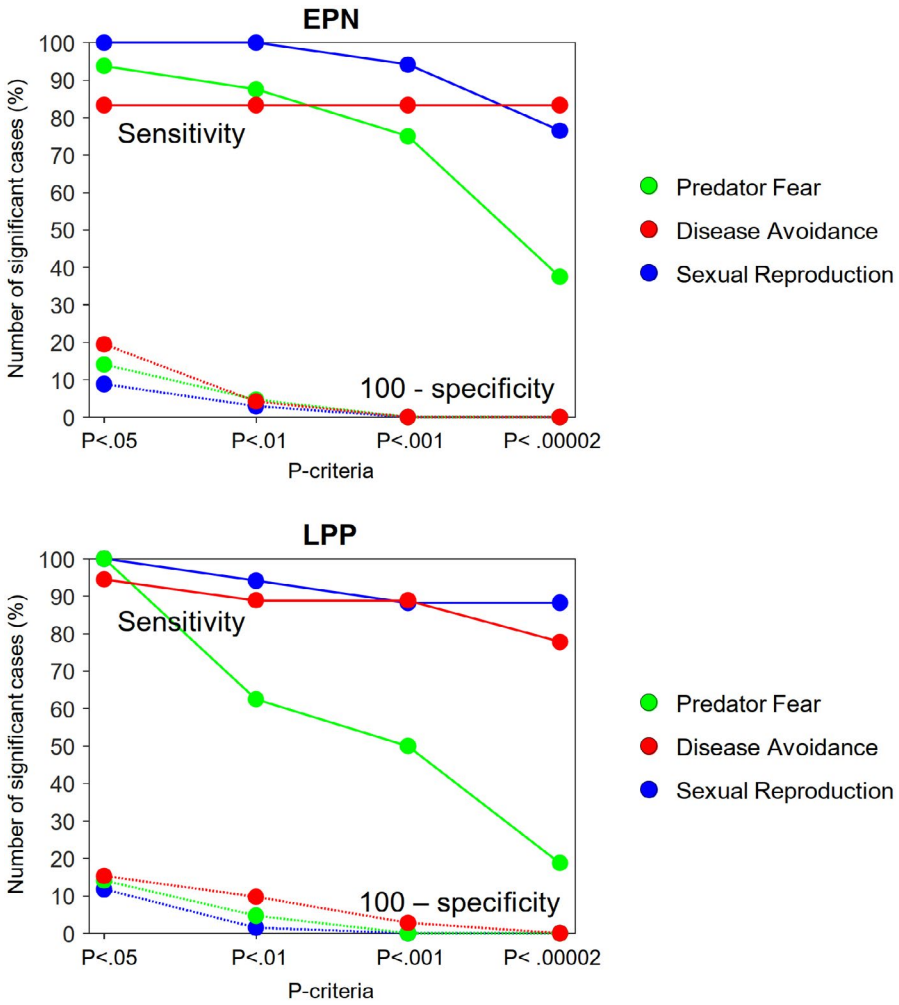

FIGURE 4 Illustration of the sensitivity and specificity analysis as a function of different p-level criteria for the EPN (upper panel) and LPP (lower panel) components. The y-axis displays the percentage of cases showing significant effects in the sensitivity analysis and the percentage of cases showing false positive effects in the specificity analysis (false positive rate: 100-specificity)

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for the EPN and LPP components as used for the group analyses. Relying on a $p < .05$ criteria, the majority of participants showed a significant emotional modulation effect for the EPN (14/16, 15/18, and 17/17) and LPP (13/16, 15/18, and 17/17) for predator fear, disease avoidance, and sexual reproduction. Furthermore, regarding stronger *p*-level criteria for determining statistical significance, a similar pattern of findings as observed for the main analysis emerged (see Figure 5, upper panels).

Further analyses explored the robustness of single subject analysis focusing on the length of the time window used to score the components. Toward this end, the time window used for scoring was doubled and the EPN and LPP scored in time windows of 120 and 200 ms duration, respectively. With regard to the $p < .05$ criteria, findings for disease avoidance and sexual reproduction were similar for EPN (15/18, and 16/17) and LPP (16/18, and 16/17) as reported in the main analysis (see Figure 5, lower panels). However, for predator fear, doubling the time window led to a reduced number of significant EPN (13/16) and LPP (11/16) effects.

4 | DISCUSSION

This research series was based on consistent and reliable findings in group research showing that the processing of high compared to low arousing emotional stimuli is associated with increased EPN and LPP components (Hajcak et al., 2010; Schupp, Flaisch, et al., 2006). However, general principles of emotion processing can not only be derived with reference to group data but also at the level of the individual case. Specifically, the case-by-case approach was used here to explore the hypothesis that the emotional modulation of the EPN and LPP components can be revealed at the level of the individual case. Across three studies exploring behavior systems of predator fear, disease avoidance, and sexual

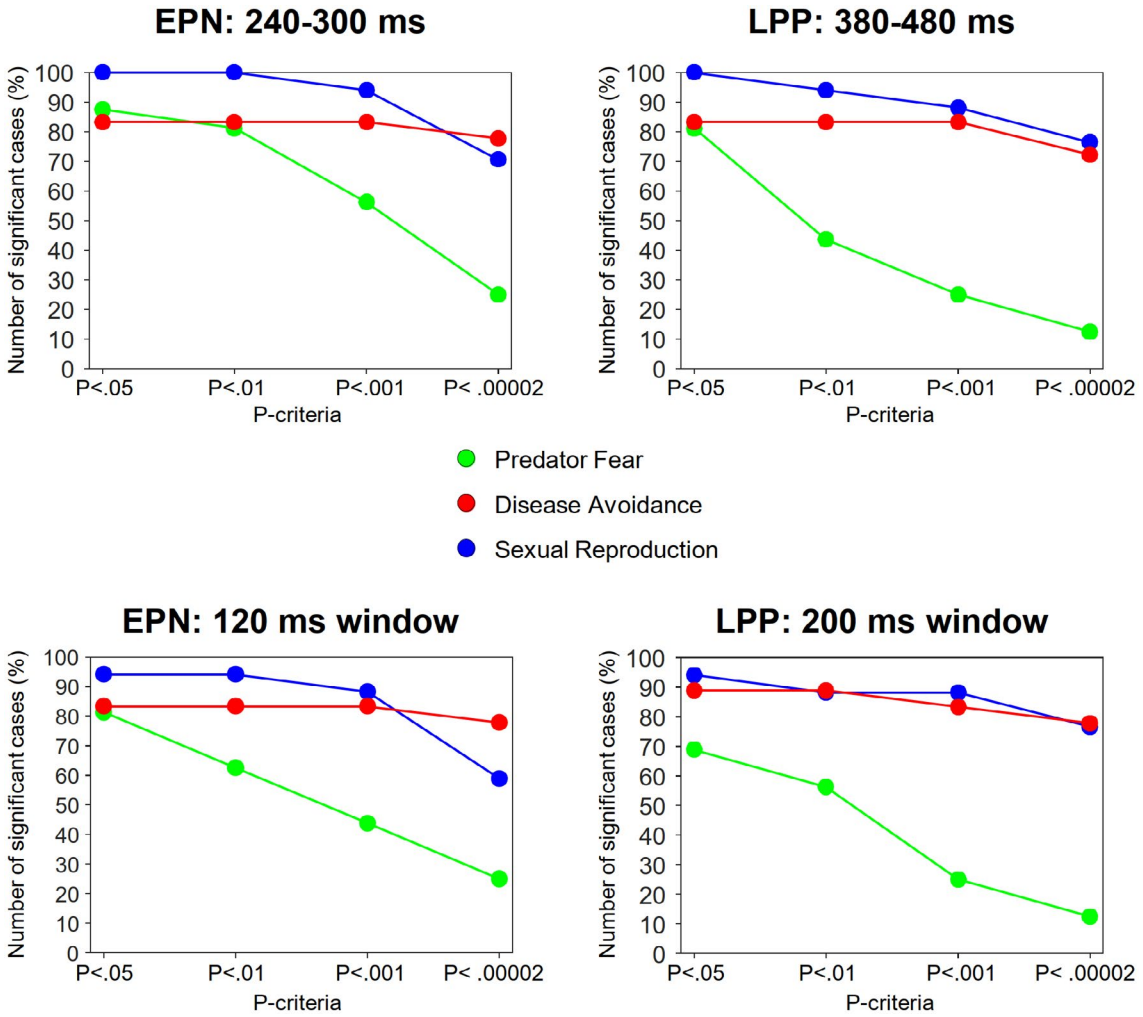

FIGURE 5 Illustration of the outcome of the control analyses conducted using different approaches to score the EPN and LPP components. Upper panel: In these analyses, the EPN and LPP components were scored with the same sensor cluster and latency windows as used in the group analysis. Lower panel: These control analyses show the findings obtained when using extended time windows to score the EPN and LPP. The *y*-axis displays the percentage of cases showing significant emotional modulation effects

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reproduction, 47 and 50 out of 51 individual tests indicated significantly larger EPN (92%) and LPP (98%) amplitudes to emotional stimuli high in arousal. Considering direct replication within each of the three studies and systematic replication across studies (Sidman, 1960), the findings suggest the case-wise reproducibility of the emotional modulation of the EPN and LPP components.

The principle of emotion processing investigated here regards the process of affective stimulus evaluation. The organism encounters many stimuli in the environment which are evaluated according to their emotional meaning and significance. The phenomenon has been captured by Frijda in his law on situational meaning, proposing that "emotions arise in response to the meaning structure of given situations" (Frijda, 1988, p. 349). Similarly, Asch has pointed out that "Our hopes and despairs seem to spring directly from the sensed significance of events." (Asch, 1952, p. 110). The network model of emotion provides an integrative account by proposing that emotional responding is consequent upon the activation of a matching long-term memory representation that includes sensory-perceptual, conceptual-meaning, and response-action nodes (Lang, 1993). Furthermore, the notion of motivated attention suggests that the activation of these emotional networks provides an important signal to direct selective attention processes (Lang et al., 1997). Building upon the similarities of explicit and intrinsic stimulus significance, it has been suggested that the EPN and LPP components relate to the emotion–attention relationship signifying the emotional significance of stimuli (Lang & Bradley, 2010; Schupp, Flaisch, et al., 2006). Given that EPN and LPP effects were observed for behavior systems of approach (sexual reproduction) and avoidance (predator fear and disease avoidance), these components relate to emotional arousal rather than the valence of the stimuli. The main contribution of the present research series regards the unit of analysis to reveal this principle of emotion functioning. Being firmly established in group research (Hajcak et al., 2010; Schindler & Bublatzky, 2020; Schupp, Flaisch, et al., 2006), the present data reveal the phenomenon at the level of the individual case. While awaiting replication in future research, the present findings appear promising that these two ERP components can be useful to probe the process of affective stimulus evaluation at the level of the individual case. Furthermore, while the focus here was on initial stimulus processing, future research may utilize experimental paradigms suited to assess the sustained processing of emotional pictures.

Determining a general principle of emotion processing with brain measures needs to consider the issue of approximate replication (Kapur et al., 2012). Specifically, neuroscientific studies generate large data sets with multiple possibilities for data analysis. In group research, approximate replication refers to the acceptance of findings as a replication although they may not reflect the same effect. The concern of approximate replication is also relevant for the case-by-case approach. Here, EPN and LPP sensor clusters were derived from group analysis and the same sensor clusters used for the analysis of each study, emotion category, and individual case. This approach seems rather conservative given that variations between persons in brain structure and function are ignored (Zilles & Amunts, 2010), which may lead to different ERP scalp topographies. However, the effect was allowed to vary in time within the usual time window of the EPN and LPP. Complementary, a control analysis scored the EPN and LPP component in the same time windows as used in the group analyses to provide the results for the most restrictive approach of ERP scoring. Of note, the effect of significantly larger EPN and LPP amplitudes for strong rather than weak emotional stimuli were observed in 46 (90%) and 45 (88%) out of 51 individual cases. These data support the notion of a common effect across individuals when defined in a most conservative way, that is, allowing no differences in topography and latency across individuals and emotion categories.

In addition, robustness of findings can be estimated by exploring the effects of comparatively shorter or longer time windows used to the score the ERP components. Accordingly, further control analyses were performed based on doubled time windows of the EPN and LPP components, that is, 120 and 200 ms, respectively. Findings were similar to the main analysis for behavior systems of disease avoidance (EPN: 83%; LPP: 89%) and sexual reproduction (EPN: 94%; LPP: 94%). A differential pattern of results emerged for predator fear. Specifically, the number of significant cases here was reduced (EPN: 81%; LPP: 69%), and strongly declined with respect to using more strict p-criteria to determine significant effects. Assessing for each case the processing of multiple rather than a single emotion category appears as a promising avenue to further assess effects of different scoring approaches in future research.

Reliance on the $p < .05$ criteria for determining strong and robust effects has been questioned for the evaluation of study findings in group-based research (Kapur et al., 2012). The issue is also pertinent to the case-by-case approach, in particular regarding the interpretation of individual test results and the possibility to translate basic science findings to applied settings. Considering the p-value as a graded measure of evidence against the null hypothesis (Button et al., 2013; Halsey et al., 2015), Figure 3 displays the proportion of participants showing a significant EPN and LPP effect as a function of different p-value levels. Pronounced differences for the three emotional stimulus categories are seen. Specifically, categories of sexual reproduction and disease avoidance result in high numbers of significant cases for both ERP components, that is, between 76% and 88%, even under the most stringent p-criteria ($p < .00002$). This is in stark contrast with the findings for the predator fear category, which shows a marked decline of significant

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effects with increasing p-criteria. Furthermore, considering the findings of the specificity analysis, reliance on the conventional $p < .05$ criteria can lead to false positive findings. Accordingly, depending on the specific research goals, more stringent p-level criteria, that is, $p < .01$ and beyond, may be applied to increase the specificity of the analysis. Overall, considering both sensitivity as well as specificity, categories of sexual reproduction, and disease avoidance appear as prime stimulus categories to reveal the emotional modulation of the EPN and LPP components at the level of the individual case.

While the findings appear promising, some limitations need to be acknowledged. Specifically, as Studies 1–3 included college students, the generalizability of the findings needs to be shown in representative samples. Furthermore, Study 3 examining stimuli related to sexual reproduction included only female participants. While previous group research suggests stronger responses to erotic stimuli in male than female participants with regard to brain activity in extrastriate cortex (Sabatinelli et al., 2004), amygdala and hypothalamus (Hamann et al., 2004), and peripheral physiological responses (Bradley et al., 2001), future research is needed to conduct case-by-case analyses to erotic stimulus processing in male participants. In addition, robustness and reproducibility of the present research can and should be further improved within a systematic research program (Bradley, 2017). Improvements in the selection of stimulus materials provide one promising avenue. Specifically, establishing stimulus categories producing strong effects may serve as foundation for the study of stimulus materials idiosyncratic to the individual case. Furthermore, future research determining the stability of the effects across time seems informative with regard to possible contributions of a case-by-case approach to biomarker development and translation to the clinical domain. Selective attention deficits to significant stimuli have been observed in mental disorders, involving for instance the anxiety spectrum, eating disorders, and drug addiction (De Houwer et al., 2004; Mogg & Bradley, 1998; Shafran et al., 2007). Furthermore, the Research Domain of Criteria (RDoC) project by the National Institute of Mental Health aims to identify new ways of classifying mental disorders, in which groups of individual patients are clustered based on their individual response profile (Cuthbert, 2014; Insel & Cuthbert, 2015). While beyond the aim of the present research, the cause-effect relationship on emotion functioning examined here may be potentially of clinical utility.

One implication of these findings is to consider the case-by-case approach as a further strategy to assure reproducibility of findings in neuroscience. The series of replication studies in psychological science in recent years has cast doubt on the reproducibility of study findings (Open Science Collaboration, 2015), and a similar state of affairs has been indicated in the field of neuroscience (Button et al., 2013; Ioannidis, 2005; Poldrack et al., 2017). Important suggestions for improvements have been made, that is, larger samples, more replication studies, and the pre-registration of experimental protocols. However, with the group study as default mode, the case-by-case approach has received little attention yet. The present findings demonstrate that the case-by-case approach can be useful for revealing robust and replicable empirical regularities (Danziger, 1990; Lamiell, 1998).

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AUTHOR CONTRIBUTIONS

Harald Schupp: Conceptualization; Formal analysis; Funding acquisition; Investigation; Methodology; Visualization; Writing-original draft. **Ursula Maria Kirmse:** Formal analysis; Investigation; Methodology; Software; Visualization; Writing-review & editing.

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